

modules. "If we have a wireless infotainment system, we need to test it for wireless standardisations as well," says Kalyan Verma, vice president, products, TUV Rheinland India. Test solutions would typically require maximum radio frequency (RF) testing performance with flexible chip control through several software layers.

With autonomous vehicles making rapid paces in recent times, advanced driver-assistance systems (ADASes) have been on the rise. ADASes today are about taking control of the vehicle away from the driver. Sensor technologies like radars, ultrasounds, cameras and dedicated short range communication for vehicle to everything (DSRC for V2X) present attack surfaces that could seriously impact the safety of the vehicle. Radar or DSRC signals can also be easily spoofed, confusing ADAS algorithms into steering in the wrong direction, or driving into an obstacle.

With smartphones coming into play, another issue of security arises with software applications. A Trojan Horse computer program can easily gain entry into the infotainment system through a paired Bluetooth device or Wi-Fi connection. Media content played via USB and CD are issues along similar lines. Vulnerabilities in the media code can be exploited to plant malicious software in the infotainment system. Weak information technology security further adds to the problems.

Case study

HARMAN/Becker Automotive Systems and Noffz Technologies created a scalable and reusable test station. Requirement for shorter test times per device under test (DUT) combined with a higher throughput in general was achieved by RF instrument sharing or multi-DUT testing. The resulting system allowed the testing of four DUTs with four ports, each including the cellular main and diversity antenna, GNSS and Bluetooth/WLAN.

EQUIPMENT FOR VARIOUS PRODUCTS

Name of product	Testing done
NCS Titan by IFEN	GNSS simulators
NI PXI platform	Infotainment systems
Automotive tool box from Microlease	Infotainment systems
ADAS radar test solution from National Instruments	RF measurements and target simulation for radar sensors
RSAT100A signal analyser from Tektronix	Wide-band radar and electronic warfare systems
MB040A from Keysight	Physical-layer characterisation and compliance testing
Modular automotive simulator from Cruden	ADAS, anti-virus and HMI testing
AST-1000 RF signal source from Avera	Automotive infotainment systems
UTP 9010 series from Noffz Technologies	Functional test, infotainment test and RF test
Rohde & Schwarz ESW EMI test receiver	Electromagnetic compatibility tests

Increasing number of modules in the infotainment system. Modular testing systems have been a big help in this regard. PXI platform from National Instruments is one solution for modular hardware platforms that can be configured with key software tools. With traditional approach to automotive infotainment RF tuner testing, test engineers need multiple instruments to generate different broadcast and navigation standards. Modular systems allow testing of multiple parameters using the same instruments. For example, the same signal vector generator can be used to simulate GPS and digital video broadcast by simply reprogramming PXI platform.

"Human-machine interface/user interface (HMI/UI), infotainment features or functionalities, common tests and stress tests are the four major perspectives with current infotainment platform validation," says Deepak Hebbur, co-founder, Skillfinity. He adds, "The features are tested as per requirements and functionality." Common tests are for quality, response time and power consumption. Stress tests include voltage curves, electromagnetic compatibility and heat emissions.

Traditional RF tuner testing required multiple instruments to generate different broadcast standards. Multiple instruments were required to generate multiple carrier signals to test the critical functionalities in

the infotainment systems, including channel search, alternate frequency selection, emergency and traffic announcements, among others.

Multiple high-definition acquisitions and synchronisation is required to detect issues arising from such systems. Surround sound, encryption schemes and full-HD video in latest infotainment systems have made multimedia test applications increasingly more complex. One example of automotive test that requires an integrated, high-performance solution is ensuring the synchronisation of multiple screens for an infotainment system. Hebbur says, "Validations are also done to confirm the robustness of the UI where you try multiple button presses in a crazy sequence to check if the system handles all the presses gracefully."

Testing electronics

All electronics in a vehicle must do their tasks properly. Verma says, "Different modules are supposed to work as an environment and, hence, call for testing at various stages."

"For proper autonomous vehicles, we work on individual modules, testing these against the required parameters. After module testing, the environment as a whole is tested," adds Gupta. "Lab car is used to simulate the vehicle's running environments, and parameters are measured to check its proper